Hardware Accelerator for YOLOv5 Object Detection on FPGA

using N:M Structured Sparsity Pruning

HCMUTE · Faculty of Advanced Education | Vu Tien Dat (22119006) · Nguyen Tran Tien (22119030) | Advisor: Assoc. Prof. Dr. Vo Minh Huan | July 2026

01 INTRODUCTION & MOTIVATION

- Edge detection needs accuracy + low latency + low power at once — FPGA delivers all three at low wattage.
- Embedded GPU 10–15 W (Jetson) · ARM CPU too slow for CNNs → FPGA is the balanced middle ground.

Objectives: ① YOLOv5s-Lite for edge ② ≥10× compression ③ unified RTL + native N:M decode ④ bit-exact on KV260.

THE GAP: YOLOv5s needs 7.2M params / ~7 GFLOP @416 — far beyond KV260 (117K LUT · 1,248 DSP · 64 URAM)

02 THREE-STAGE MODEL COMPRESSION

- YOLOv5s → Lite (C3_DWS, depth=1, LeakyReLU) → INT8 QAT → N:M pruning.

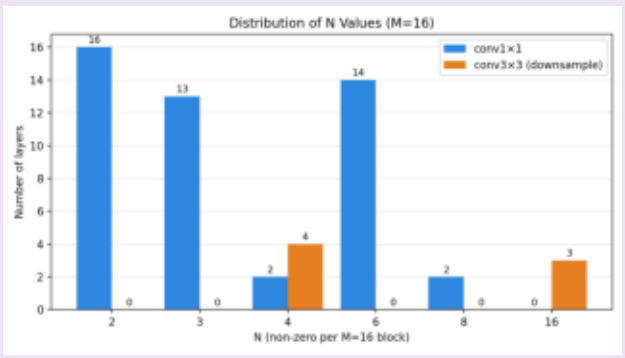
Stage	Params	FLOPs	Wt	mAP
YOLOv5s (base)	7.2 M	~7 G	14.4M	—
+ YOLOv5s-Lite	5.15M	4.68G	2.9 M	.9945
+ INT8 QAT	5.15M	4.68G	1.28M	.9943
+ N:M (71.9%)	5.15M*	1.31G	0.36M	.987

* 5.15M stored ≈ 1.16M non-zero · N:M eff 1.31G (dense 4.68G).

- ≈40× weight compression · 98.7% mAP kept · arch ~1.4×/~1.5×.

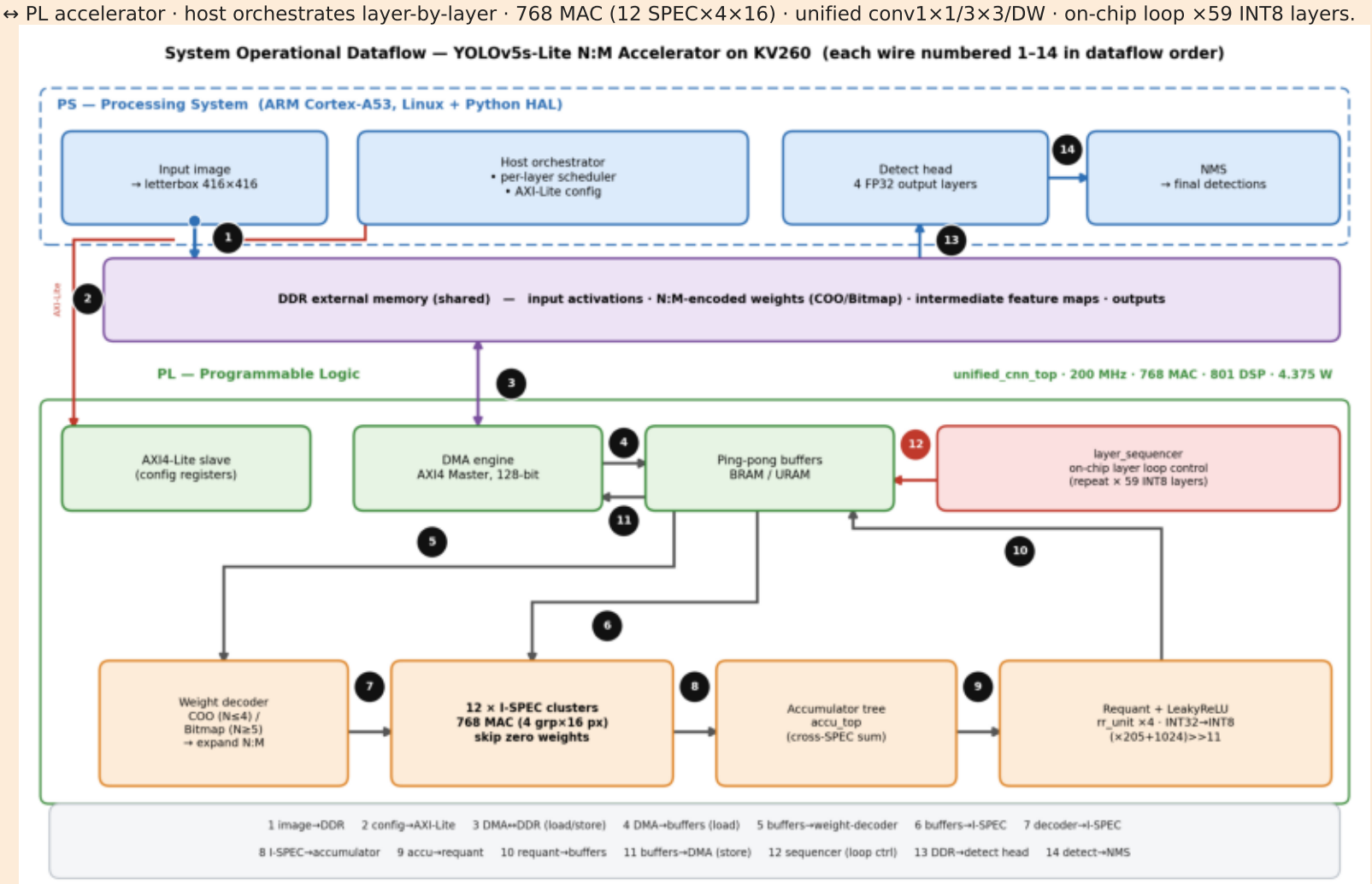
03 N:M SPARSITY + DominoSearch

- Keep the N largest of every M=16 weights → sparsity = (M−N)/M.
- DominoSearch: init dense N=16 → greedy η-descent lowers N.
- Fragile stay N=16 · robust → N=2 · COO (N≤4)/Bitmap (N≥5).



04 SYSTEM ARCHITECTURE ★

- PS (ARM) ↔ DDR ↔ PL accelerator · host orchestrates layer-by-layer · 768 MAC (12 SPEC×4×16) · unified conv1×1/3×3/DW · on-chip loop ×59 INT8 layers.



05 RESULTS ★

200 MHz · 4.375 W · ~94 GOPS/W · 1.30 FPS

LUT 62.9% · DSP 64.2% (801) · BRAM 93.8% · URAM 87.5% · 768 MAC

100% bit-exact (59/59) · FPGA ≡ CPU-INT8 (IoU 1.00)

CPU IoU: dense→pruned 0.62 · pruned→INT8 0.76



06 CONCLUSION & FUTURE WORK

Achieved:

✓ YOLOv5s-Lite 5.15M · mAP 0.9945 · ≈40× compression · 98.7% kept.

✓ Unified RTL · 100% bit-exact · 200 MHz · WNS +0.004 ns.

Future work:

→ Double-buffering (cut DMA idle 30–50%) · multi-class (COCO/VOC).

→ Migrate to YOLOv8 · ASIC for higher energy efficiency.